

- i. Binary to Decimal: A = 00100001
- ii. Binary to Hexadecimal: B = 11000011
- iii. Binary to Decimal: C = 1000.01
- iv. Two's Complement of D: D = 00100001
- v. Signed binary representation of E: E = -33 [read: negative 33]

Decimal of A =

Hexadecimal of B =

Decimal of C =

Two's Complement of D =

Signed binary of E:

7. [10 points]

7.a [6points] Complete the state table of Table 2 and determine the CSOP next state and output expressions corresponding to the state diagram of Figure 3. Give the CSOP expressions in minterm list form. The state codings are to the right of the state name in the states. Make any assumption necessary about any unused states.

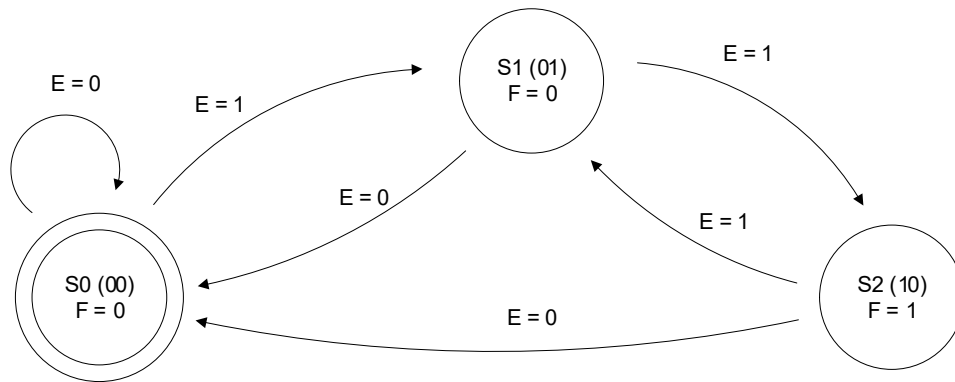


Figure for Problem #7. Sequential Circuit State Diagram.

Q1 Q0	E	F	Q1 ⁺ Q0 ⁺
0 0	0		
0 0	1		
0 1	0		
0 1	1		
1 0	0		
1 0	1		
1 1	0		
1 1	1		

State Table for the Sequential Circuit of Problem #7.

[2 points]

Q1+ =

[2 points]

Q0+ =

[2 points]

F =

7.b [4 points]

[2 points each] Complete the timing diagrams enclosed below corresponding to a single D flip-flop as shown in the Figure for Problem 7.b. Name and use the two lines indicated next to Q as the levels needed for Vcc/1 and GND/0, and use vertical lines to indicate all clock edges used.

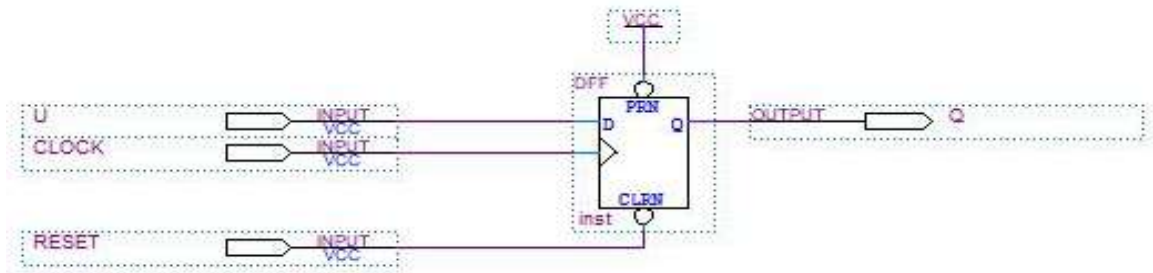
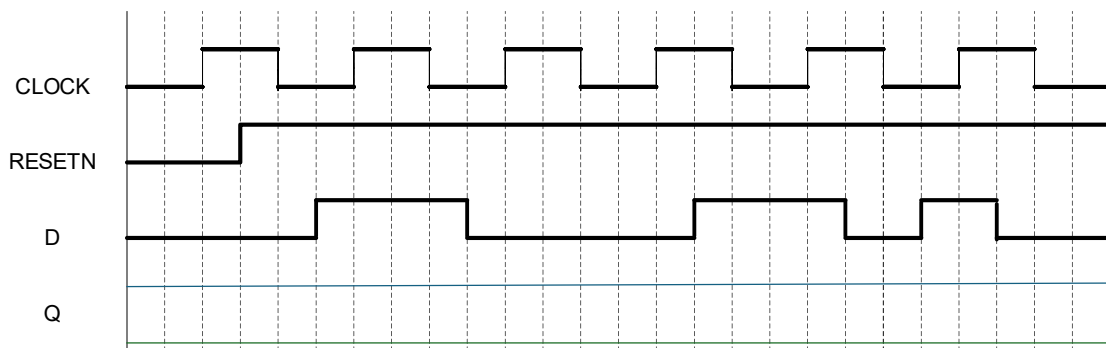
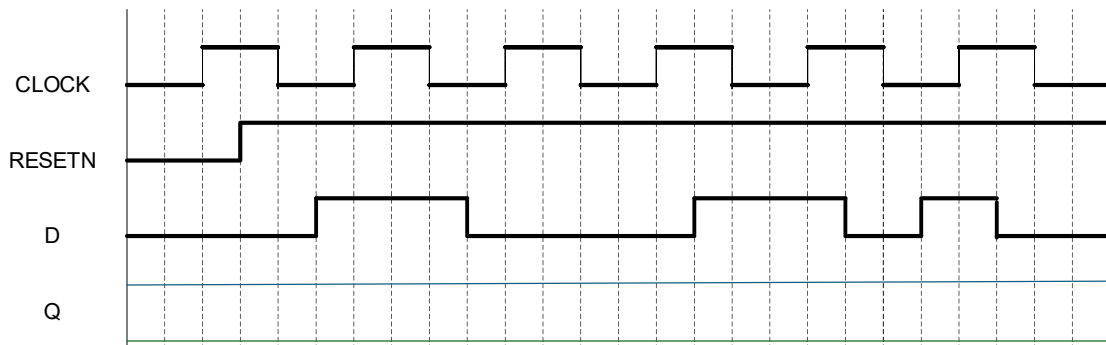


Figure for problem 7.b: D Flip-flop Circuit



Output Q at the ***positive*** edge of the clock. Mark the edge with an arrow.



Output Q at the ***negative*** edge of the clock. Mark the edge with an arrow.

9. [10 points] Provide a complete design for a sequence recognizer using D Flip-Flops, as follows:

All sequences/substrings occurrences of: **101** in a continuous bit stream, using $U(t)$ and $F(t)$ below

Input Sequence: $U(t)$: 0 1 0 1 0 0 1 0 1 0 1 1 0 0 1 0 1 ...

Output: $F(t)$: ? ? 0 1 0 0 0 0 1 0 1 0 0 0 0 0 1 ...

[4 points: state diagram/state assignment; 4 points: state transition table, and 2 points: flip-flop input/next state equations and output equations]

10. [10 points] You **must** use a powers-of-2 format for your answer as applicable to the following questions.

10.i [5 points] For a memory system with a 30-bit address bus and 16-bit data bus:
[one point each of the following five questions]

a) How many memory addresses (locations) does the memory system have?

Addr =

b) What is the word size of the memory?

Data =

c) What is the memory capacity; in both bits and Bytes

M [bits] =

M [Bytes] =

d) What additional signals (two of them) are typically needed to control a memory system?

e) How much more memory capacity would the memory system specified above have if the address bus is bigger by **one bit**; that is 21 bits instead of 20 bits.

10.ii [5 points] Specify the register transfer operations in RTL for the following digital system operations (your answers should be using RTL **not** microcode):
[one point each of the following five questions]

a) Add the contents of registers 10 and 11 and place the result in register 8.

b) Store the contents of register 15 in the memory location referenced by the contents of register 30.

c) Multiply the value stored in register 20 by 4 and store it back in register 20.

Specify the datapath microcode to perform the operations specified by the following register transfer operations.

d) $R15 \leftarrow R10 - R11$

e) $R8 \leftarrow M[R27]$

Signed Statement

I neither received nor gave any aid on any problem on this exam. I did not use any resources other than those specified.

Print Name: _____

Sign Name: _____ Date Signed: _____